

Course Description

Module Title

English title: Natural Language Processing and Network Analysis

Danish title: Natural Language Processing og netværksanalyse

ECTS

Number of ECTS: 5

Location

Education programme: Master of Science in Economics and Business Administration (Business Data Science)

Semester: 1st

Module Coordinator

Module coordinator: Roman Jurowetzki

Type and Language

Module type: Course

Language of instruction: English

Learning Objectives

KNOWLEDGE

The objective is that the student after the module possesses the necessary knowledge on:

- the conceptual particularities and explanatory power of relational and network data as well as unstructured and text data including understanding of the epistemology of relational and language data.
- the interplay between network-theory concepts and real-world networks as well as the representation and analysis of real-world phenomena from unstructured data and text
- the integration of relational and unstructured, particularly language data into statistical and machine learning based methodologies as well as the interpretation of findings in business and economics.

SKILLS

The objective is that the student after the module possesses the necessary skills in:

- sourcing, storing and pre-processing network and text data – including using various techniques of vectorisation - calculating and interpreting essential statistic metrics, and integrating network and text indicators into machine learning pipelines.
- visualising text data, networks and interaction pattern.
- performing tasks such as grammar-based labelling and modifications on text data, automated summarisation, sentiment analysis, and extracting entities from text using non-neural approaches.

COMPETENCES

The objective is that the student after the module possesses the necessary competences in:

- applying a problem-based approach to central challenges within natural language processing and network analyses including representing any real-life complex systems as networks.
- identifying latent patterns, structures and interactions of entities in these systems, and exploring the interplay between the structure of systems and their performance as well as particular features and behaviour of individual entities.
- utilising natural language data for various types of mapping and analysis.

Academic Content and Conjunction With Other Modules/Semesters

This module aims to give students insight into network and unstructured data types such as natural language and text, as well as state-of-the-art approaches to map and analyse these data. Insights and techniques gained in this module will allow students to approach real-world problems in marketing (Who are the main influencers among our customers?), management (Can we identify new discourses in the communication within our organisation?) or business economics (Can language patterns be used to understand R&D intensity across companies).

Upon completion, students will have built a solid knowledge foundation within network theory and analysis, computational linguistics and broader (unstructured) data processing. The module is application-focused, and thus students will gain a variety of skills to utilise relational and unstructured text data for analysis purposes.

Scope and Expected Performance

5 ECTS corresponds to a workload of 135 hours, distributed among the following activities:

With teacher (including quest teachers and TAs):

Lectures: 12 hours

Exercises/Seminar: 12

hours In total: 24 hours

Without teacher:

Digital lectures: 0 hours

Exercises/Casework: 27 hours

Curriculum preparation: 84 hours

In total: 111 hours

Project writing with supervision: 0 hours

1. Session: Intro to NLP: From Classic Techniques to Modern Architectures

- **Instructor:** Roman Jurowetzki
- **Format:** 2 hours lecture, 2 hours exercise
- **Description:** This foundational session charts the evolution of Natural Language Processing (NLP). We begin with the classic techniques that shaped the field, such as Bag-of-Words (BoW) and TF-IDF, which are crucial for understanding the principles of text representation. We then progress to pivotal word embedding models like Word2Vec and GloVe that first captured semantic meaning. The session culminates with an exploration of sequence-to-sequence (Seq2Seq) architectures, the direct predecessors to the Transformer models that dominate NLP today.
- **Readings:**
 - Bird, S., Klein, E., & Loper, E. (2009). *Natural language processing with Python: analyzing text with the natural language toolkit*. O'Reilly Media, Inc.

2. Session: The Transformer Era: Transfer Learning and Large Language Models (LLMs)

- **Instructor:** Roman Jurowetzki
- **Format:** 2 hours lecture, 2 hours exercise
- **Description:** This session explores the paradigm shift brought by transfer learning and Large Language Models (LLMs). We will dissect the Transformer architecture, focusing on the attention mechanism that enables it to process language with unprecedented context. The primary focus will be on practical application, treating state-of-the-art models (e.g., GPT series, Llama, Mistral) as powerful, accessible tools. Through hands-on exercises, students will learn to fine-tune these pre-trained models for custom tasks, demonstrating how to achieve cutting-edge results efficiently.
- **Readings:**
 - Goldberg, Y., & Levy, O. (2014). word2vec Explained: deriving Mikolov et al.'s negative-sampling word-embedding method. *arXiv preprint arXiv:1402.3722*.
 - Devlin, J., Chang, M. W., Lee, K., & Toutanova, K. (2018). Bert: Pre-training of deep bidirectional transformers for language understanding. *arXiv preprint arXiv:1810.04805*.

3. Session: Advanced LLM Applications: RAG, Agents, and Evaluation

- **Instructor:** Christian Nielsen, PhD Student
- **Format:** 2 hours lecture, 2 hours exercise
- **Description:** Building on our understanding of LLMs, this session dives into advanced, real-world applications through practical case studies. We will cover techniques for sophisticated information extraction and text classification. A key focus will be on Retrieval-Augmented Generation (RAG), a state-of-the-art method for grounding LLM outputs in factual, external knowledge bases.

Furthermore, we will address the critical topic of evaluation, implementing procedures like Inter-Rater Reliability (IRR) to ensure the outputs of our NLP systems are consistent and trustworthy.

- **Readings:**

- Schulhoff, S., Ilie, M., Balepur, N., Kahadze, K., Liu, A., Si, C., ... & Resnik, P. (2024). The prompt report: A systematic survey of prompting techniques. *arXiv preprint arXiv:2406.06608*.

4. Session: Introduction to Graphs and Network Analysis

- **Instructor:** Hamid Bekamiri
- **Format:** 2 hours lecture, 2 hours exercise
- **Description:** This session introduces the fundamental concepts of graph theory and network analysis. Students will learn about different graph types (directed, undirected, weighted), core properties (nodes, edges, degrees), and key metrics for analyzing network structure (e.g., centrality, community detection). The session will cover real-world applications in social networks, supply chains, and recommendation systems. This knowledge provides the foundation for understanding how relationships in data can be structured and leveraged in advanced AI systems.

- **Readings:**

- **Borgatti, S. P., Mehra, A., Brass, D. J., & Labianca, G. (2009).** *Network analysis in the social sciences*. Science, 323(5916), 892-895.
- **Watts, D. J., & Strogatz, S. H. (1998).** *Collective dynamics of 'small-world' networks*. Nature, 393(6684), 440-442.

5. Session: The Synergy of Networks and LLMs: Building Knowledge Graphs

- **Instructor:** TBC
- **Format:** 2 hours lecture, 2 hours exercise
- **Description:** This session explores the powerful synergy between Large Language Models and network structures, with a specific focus on Knowledge Graphs (KGs). We will investigate how LLMs can be used to automate the construction and enrichment of KGs from unstructured text. Conversely, we will explore how structured KGs can be used to ground LLM responses, enhance their reasoning capabilities, and reduce factual inaccuracies. This session bridges the gap between probabilistic language models and symbolic knowledge representation, a key frontier in modern AI.
- **Readings:**
 - **Pan, S., et al. (2024).** *Unifying Large Language Models and Knowledge Graphs: A Roadmap*. (IEEE Transactions on Knowledge and Data Engineering)
 - **Edge, D., et al. (2024).** *From Local to Global: A Graph RAG Approach to Query-Focused Summarization*. (Microsoft Research)

6. Session: Workshop: Prototyping an Integrated AI Application

- **Instructor:** TBC
- **Format:** 2 hours lecture, 2 hours exercise
- **Description:** This capstone session is a hands-on workshop where students will integrate the concepts from the entire module to build a sophisticated NLP application. Participants will work on a project that combines the power of Large Language Models (from Sessions 2-3) with the

structured insights from Network Analysis and Knowledge Graphs (from Sessions 4-5). The goal is to prototype an end-to-end solution, demonstrating the ability to design, build, and present a modern, multi-component AI system.

A curated collection of 3-5 current articles and case studies will be provided to illustrate the application of NLP, Networks and Data Science in real-world business scenarios.

Examination

Name of the exam: Natural Language Processing and Network Analysis

Type of exam: Oral exam

Individual examination.

Assessment: 7-point grading scale

Type of grading: Internal examination

Prerequisite for participation in the exam:

- A prerequisite for participating in the exam is that the student has handed in written material.

How is the exam conducted:

Students will receive a question/problem to prepare – with 30 minutes of preparation time before the exam. They will have access to some materials provided along with the question. The examination will start with a presentation of the prepared question, followed by a Q&A session covering the course curriculum and material submitted throughout the module. During the examination, the student will have access to the materials used during the 30-minute preparation period (scope and amount specified with the question/problem) as well as the assignment/portfolio material submitted by the student prior to the exam.

Allowed aids: A computer with internet access may be used to demonstrate non-written elements of the project assignment (e.g., apps, computational notebooks, code). Otherwise, only minimal written notes (max. 10 pages) and a printed or on-screen PDF version of the project are permitted.

Communication with others and the use of generative AI tools during the examination are not allowed, except when demonstrating functionality that was developed as part of the project.

Information about the re-exam:

According to the examination regulations, section 1.1 on the determination of time and place for holding exams, you as students can expect that information about the exam form, time, place, and permitted aids will be communicated no later than 2 weeks before the re-exam is held. If the re-exam is conducted digitally, this will also be communicated no later than 2 weeks before the re-exam is held.

For exam date and further information go to Moodle.

